

A new grafting monomer for synthesizing long chain branched polypropylene through melt radical reaction

Zhenjiang Zhang^{a,c}, Dong Wan^{a,b}, Haiping Xing^{a,b}, Zhijie Zhang^{a,b}, Haiying Tan^{a,b}, Lu Wang^{a,b}, Jun Zheng^{a,b}, Yanjie An^{a,b}, Tao Tang^{a,*}

^a State Key Laboratory of Polymer Physics and Chemistry, Changchun Institute of Applied Chemistry, Chinese Academy of Sciences, Changchun 130022, China

^b Graduate School of the Chinese Academy of Sciences, Beijing 100039, China

^c Liming Research Institute of Chemical Industry, Luoyang 471000, China

ARTICLE INFO

Article history:

Received 20 July 2011

Received in revised form

30 October 2011

Accepted 19 November 2011

Available online 25 November 2011

Keywords:

Long chain branch

Melt grafting

Polypropylene

ABSTRACT

A grafting monomer (*p*-(3-butenyl)styrene, BS) containing two carbon–carbon double-bonds with different reactivity was used in melt radical reaction to prepare long chain branched polypropylene (LCB-PP). High-temperature size-exclusion (HT-SEC) coupled with triple detectors, rheometer and NMR were used to characterize the microstructure of the resultant LCB-PP. BS showed double functions in mediating radical reaction, i.e. stabilizing macroradicals and promoting branching reaction. Compared to styrene and divinylbenzene, using BS as a grafting monomer led to the formation of more uniformly distributed LCB structure on the PP backbone. Thus the resultant sample showed strain hardening in extensional flow and had high melt strength.

© 2011 Elsevier Ltd. All rights reserved.

1. Introduction

The relationship between topological structure and properties is a charming issue in polymer science at all times [1–8]. The existence of long chain branch (LCB) on polymer backbone strongly influences the properties of polymers, such as melt properties [9–16]. For example, the melt of long chain branched polypropylene (LCB-PP) shows strain hardening in extensional flows [17–22], which is very important to polymer processing where extensional flows are dominant, such as foaming, blowing and coating. Although polymer chemistry is well developed today, it is still not easy to synthesize polymers with well-defined LCB structure, especially LCB-PP. So far several methods have been devised to produce LCB-PP based on different chemical reaction mechanism, such as in situ polymerization in a reactor [23–27], high-energy irradiation [18,20,28–30] and melt radical grafting [31–33]. The third method seems easy and simple, but it practically faces many problems due to the various reaction environments, such as dispersion state and compatibility of coagents and side reactions during melt reaction. Peroxide-initiated chain scission and coagent-assisted crosslinking are two main side reactions during the melt reaction of PP, which will result in bimodal distributions of

molecular weight and branching [34]. Various strategies have been proposed to restrain the degradation of PP during melt radical grafting, for instance, prolonging the lifetime of macroradicals by reacting with dithiocarbamate radicals reversibly [35–37] and stabilizing macroradicals by changing radical intermediate states using styrene [38,39] or furan derivatives [40,41]. Unfortunately these methods need to add another coagent except grafting monomer and have little effect on the bimodal distribution of molecular weight and branching. In recent years, various divinyl monomers containing styrenyl group and a pendent alkene group were used to synthesize well-defined copolymers with pendent alkene [42]. Dong used a divinyl monomer to synthesize star polymers bearing a vinyl-functionalized core [43]. Langston used T-agent (*p*-(3-butenyl)styrene) as both comonomer and chain transfer agent in metallocene or Ziegler–Natta catalyst system to synthesize LCB-PP [26,44].

In this work, we demonstrated a novel route to synthesize LCB-PP by melt reaction. Here a monomer (*p*-(3-butenyl)styrene, BS) containing two different kinds of carbon–carbon double-bond was used to restrain side reactions and promote branching reaction. The reactivity of the double-bond connected to benzene ring with respect to radical addition is much higher than that of the other double-bond in the BS. Their reactivity ratio is 17,000 calculated on the basis of the (*Q*, *e*) values from Alfrey–Proce theory [42]. Thus the higher reactive double-bond will preferentially react with macroradicals to stabilize macromolecules quickly, while the lower

* Corresponding author. Tel.: +86 431 85262004; fax: +86 431 85262827.
E-mail address: ttang@ciac.jl.cn (T. Tang).

reactive double-bond will later react with macroradicals to avoid the formation of hyper-branched or even crosslinked structure and promote the formation of LCB structure. Another advantage of using BS is a good compatibility between PP melt and BS. The results showed that the obtained LCB-PP had more uniform distribution of branching and molecular weight.

2. Experimental part

2.1. Materials

Polypropylene (PP, MFI = 3.2 g/10min) powder was supplied by Daqing Petrochemical Corporation, China. 2,5-Dimethyl-2,5-(tert-butylperoxy) hexane (DHBP) was obtained from Aldrich. Styrene (St) and divinylbenzene (DVB) were purchased from Beijing Chemical Factory, China. *p*-(3-butenyl)styrene (BS) was synthesized according to the literature [45].

2.2. Preparation of samples

PP samples were prepared via melt mixing at 180 °C for 7 min using an internal Brabender LH60 mixer. The screw rotating speed was 60 rpm. In order to make the agents dispersed evenly in PP powder, the peroxide and Irganox B215 were dissolved in acetone, and then the mixture and the grafting monomer were sprinkled to PP powder in a vessel. The vessel was stirred for 20 min. At last, the mixture was placed in ventilating cabinet at room temperature until the acetone was volatilized completely. When only peroxide was loaded, the resultant sample was denoted as DPP. When the grafting monomers were St and DVB, the resultant samples were denoted as PSt and PDVB, respectively. When the grafting monomers were BS, the resultant samples with various content of BS were denoted as PBS1, PBS2 and PBS3, respectively. Table 1 shows the formulation of the samples.

2.3. Characterization

Absolute molecular weight distribution and intrinsic viscosity were measured by high-temperature size-exclusion chromatography (HT-SEC) coupled with triple detectors (model: PL-GPC 220). The chromatographic system consisted of a Polymer Labs PL 220 HT-SEC with a two-angle laser light scattering (LSD) detector, a viscometer detector (VD) and a differential refractive index (DRI)

detector. The absolute molecular weight was calculated from the combination of 15° and 90° angle of light scattering detector. All experiments were carried out at 150 °C. The elution solvent was 1,2,4-trichlorobenzene, which was stabilized by 2,6-di-tert-butyl-4-methylphenol (BHT). The elution flow rate was 1.0 mL/min.

Rheological measurements were conducted for oscillatory shear and uniaxial extensional flow on an ARES rotational rheometer from TA instruments. Shear measurements were carried out in parallel plate geometry (25 mm, gap 0.8 mm) in the frequency range of 0.01–100 rad/s at 180 °C. The extensional experiments were performed at 180 °C using the extensional viscosity fixture (EVF) with extension rates between 0.01 and 0.5 s⁻¹. Melt strength was measured by using Rheotens method with a capillary of diameter of 2 mm and length/diameter of 16 at 230 °C. The melt strand was extruded with the rate of 20 mm/min and pulled down by a pair of wheels at constant acceleration of 2.4 m/min², and the force was measured till the rupture of the melt strand occurred.

Melting points (T_m) and crystallization temperature (T_c) of the samples were determined by differential scanning calorimetry (DSC) performed on a Perkin–Elmer DSC-7 instrument, using a heating rate of 10 °C/min in the temperature range 50–200 °C under nitrogen. The crystallinity of PP samples can be determined from the second heating scan using the following equations:

$$X_c = \frac{\Delta H_f}{\Delta H_f^0} \times 100\% \quad (1)$$

where ΔH_f is the specific enthalpy of melting of the samples, and ΔH_f^0 is the melt enthalpy of the hypothetically 100% crystalline PP ($\Delta H_f^0 = 209$ J/g) [46].

The gel content was determined by extraction experiment. The modified samples wrapped by quantitative filter paper were extracted in boiling xylene for 12 h. The extracted solution was charged into acetone at room temperature. The PP and the grafted PP molecules precipitated out while the unreacted grafting monomer and the homopolymer of grafting monomer remained soluble. The precipitate was filtered. The insoluble fraction and the precipitate were dried at 70 °C under vacuum for 20 h ¹H NMR spectra of the unpurified and purified modified PP samples were recorded on a Varian Unity 400 MHz instrument in tertachloroethane-*d*₂ at 110 °C.

The static mechanical properties were measured with Inston 1131 tensile testing machine, and the crosshead rate was set at 50 mm/min. For each data point, five samples were tested and the average value was taken.

3. Results and discussion

The effect of chemical structure of the grafting monomer on the molecular structures of the resulting products was studied by means of SEC coupled with triple detectors. All the modified PPs were soluble in elution solvent except PDVB and PBS3 which contained 15 wt% and 14 wt% gel, respectively (Table 1). The macromolecules could be fractionated by the SEC column and reached the three detectors to give three elution curves simultaneously. Before SEC testing, all the modified samples were purified by extraction with xylene. The responses from the three detectors (raw data) for plain PP, DPP, PSt, PBS2 and PDVB (soluble portion) samples are shown in Fig. 1. The LSD chromatogram for 90° angle (not shown) was similar to that for 15° angle, but it showed slightly lower sensitivity at higher molecular weight [47]. For plain PP, DPP and PSt, there is only one peak in the three elution curves (Fig. 1). For PBS2, there is a hump at earlier retention time in the response of LSD detector (Fig. 1c), while the hump cannot be observed in the response of RI (Fig. 1a) and IV detectors (Fig. 1b). Bear et al.

Table 1
Characterization results of plain PP and modified PP samples.

Sample	M_p (kg/mol)	M_w (kg/mol)	PDI	$[\eta]^a$ (dL/g)	g'^b	Gel (wt%)	End-torque (N m)	LCBD ^c	MS (cN)
PP ^d	219	307	3.13	1.42	1.0	0	8.8	0	2
DPP ^e	80	88	2.32	0.53	—	0	3.8	—	n.d
PSt ^f	115	137	2.15	0.68	—	0	7.2	—	n.d
PDVB ^g	126	265	4.25	0.56	0.71	15	7.5	1.95	4
PBS1 ^h	165	266	2.21	1.20	0.88	0	7.6	0.52	6
PBS2 ⁱ	176	366	2.91	1.11	0.79	0	8.9	0.83	10
PBS3 ^j	207	412	3.60	1.22	0.80	14	10.0	0.68	14

^a Weight average intrinsic viscosity.

^b Weight average branching factor $g' = [\eta]_B/[\eta]_L$, the ratio of branched intrinsic viscosity to linear intrinsic viscosity at the same average molecular weight.

^c Long chain branching points per 10,000 carbon.

^d [B215] = 2 g/kg.

^e [DHBP] = 3.4 mmol/kg, [B215] = 2 g/kg.

^f [DHBP] = 3.4 mmol/kg, [St] = 88 mmol/kg, [B215] = 2 g/kg.

^g [DHBP] = 3.4 mmol/kg, [DVB] = 88 mmol/kg, [B215] = 2 g/kg.

^h [DHBP] = 3.4 mmol/kg, [BS] = 50 mmol/kg, [B215] = 2 g/kg.

ⁱ [DHBP] = 3.4 mmol/kg, [BS] = 88 mmol/kg, [B215] = 2 g/kg.

^j [DHBP] = 3.4 mmol/kg, [BS] = 150 mmol/kg, [B215] = 2 g/kg.

observed the same phenomena when researching branched polyethylene [47]. In the case of PDVB, there is an obvious second peak in the response of LSD detector (Fig. 1c), while the second peak is small in the responses of RI (Fig. 1a) and IV detectors (Fig. 1b) especially RI response. The responses of the three detectors for plain PP and modified samples with different BS concentration are shown in Fig. 2. Four samples presented one peak in RI elution curve. However the IV elution peak was widened with increasing BS concentration and a hump appeared at earlier retention time in LS elution curve when the content of BS exceeded 88 mmol/kg.

According to the results of Yau [48], the relative intensity differences of the three detectors signals indicated the presence of LCB structure in the high molecular weight regions of PDVB and the samples modified with BS.

A more informative plot of $\log M$ versus retention time is shown in Fig. 3a. The molecular weight was calculated from the response of LSD. For plain PP, DPP and PSt, the curves of $\log M$ versus retention time are straight lines with almost the same slope. For PBS2 and PDVB, the upturn slope in the short retention time (<17.7 min) is indicative of a different kind of molecular

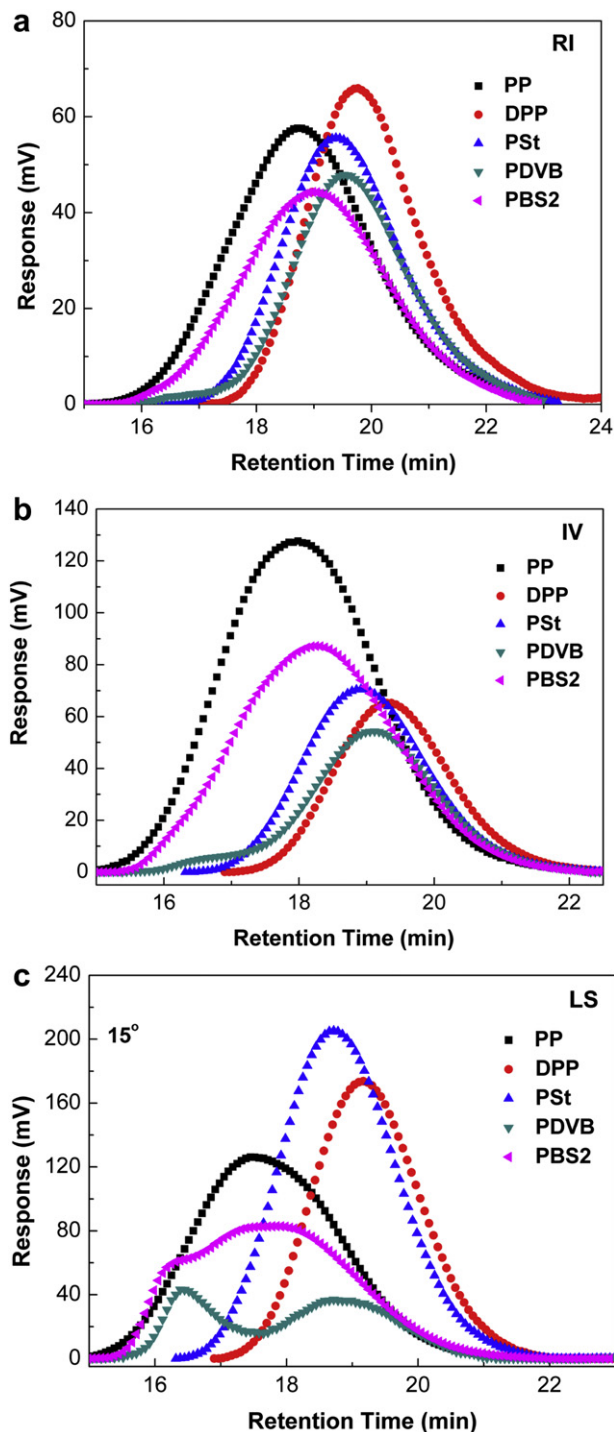


Fig. 1. Chromatograms of plain PP, DPP, PSt, PDVB and PBS2 obtained by (a) refractive index, (b) viscometer, and (c) light scattering detectors (15°).

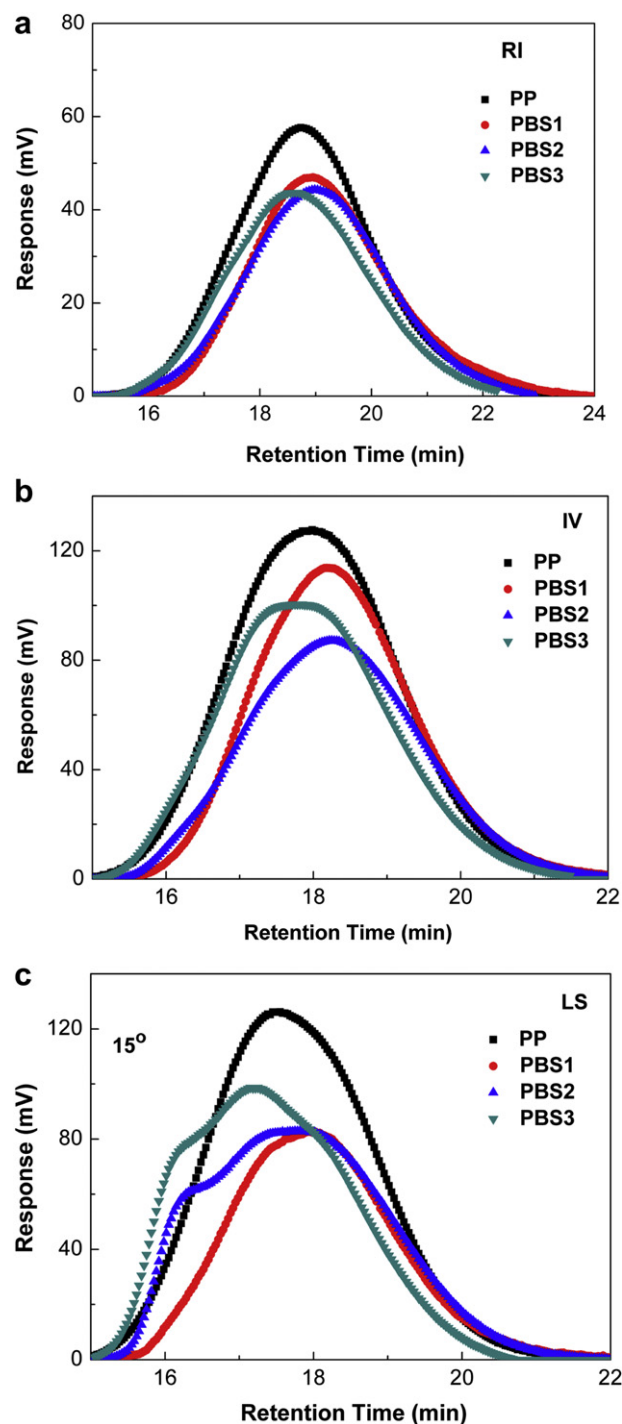


Fig. 2. Chromatograms of plain PP, PBS1, PBS2 and PBS3 obtained by (a) refractive index, (b) viscometer, and (c) light scattering detectors (15°).

populations. Fig. 3b shows log M versus retention time of plain and PBS1, PBS2 and PBS3. In the longer retention time (>17.7 min), all the samples almost had the same molecular weight at the same retention time. But the modified samples with BS had higher molecular weight at the same retention time comparing with plain PP in the shorter retention (<17.7 min). The higher than expected molecular weights in the retention shorter time are attributed to the presence of LCB. In this fraction, the higher concentration of LCB led to the higher molecular weight. Thus, the PDVB sample had a higher concentration of LCB than the samples modified with BS [47].

Fig. 4a shows the Mark–Houwink plots of plain PP, DPP, PSt, PBS2 and PDVB, which can provide the information about LCB distribution across polymer fractions. The plot of plain PP in low molecular weight region ($M < 10^6$) was a straight line with the slope of 0.705, and it deviated a little in high molecular weight region ($M > 10^6$) due to experimental error, resulting from the very low sample concentration. A similar behavior was observed in the DPP and PSt. The Mark–Houwink plots of DPP and PSt are straight line with the slope of 0.665, which indicate a linear structure. Compared to plain PP, the Mark–Houwink plots of PBS2 and PDVB shifted down in low molecular weight region. This may result from the existence of short chain branches (SCB). The SCB was probably formed due to the presence of unreacted lower reactive double-bond (PBS2) or the degradation of the long chain branches during the process of radical reactions (PBS2, PDVB). However, the SCB

does not affect the slope of the curve [9,49]. In high molecular weight region, the negative deviation of PDVB was more obvious than PBS2 in the Mark–Houwink plot, indicating that the PDVB sample has a higher concentration of LCB than that of PBS2 in high molecular weight region. The concentration of LCB can be expressed by long chain branching density (LCBD). The critical molecular weight, M_c , between long branched molecules and linear molecules can be obtained from locating the intersection of linear and branched behavior. For the samples modified with BS, M_c is $\sim 500,000$ g/mol, which is several times larger than M_p . The value of M_c suggested that most of the long chain branches existed in the high molecular weight fraction. The amount of LCB can be determined from the mean square radius of gyration $\langle R_g^2 \rangle$ and intrinsic viscosity $[\eta]$. The fundament for the branching analysis was established by Zimm and Stockmayer [50]. Based on the fact that branched chains in solution are more dense and have lower R_g and $[\eta]$ than linear chains of the same molecular weight. This is expressed through the branching factors

$$g = \frac{\langle R_g^2 \rangle_{br}}{\langle R_g^2 \rangle_{lin}} \quad (2)$$

$$g' = \frac{[\eta]_{br}}{[\eta]_{lin}} \quad (3)$$

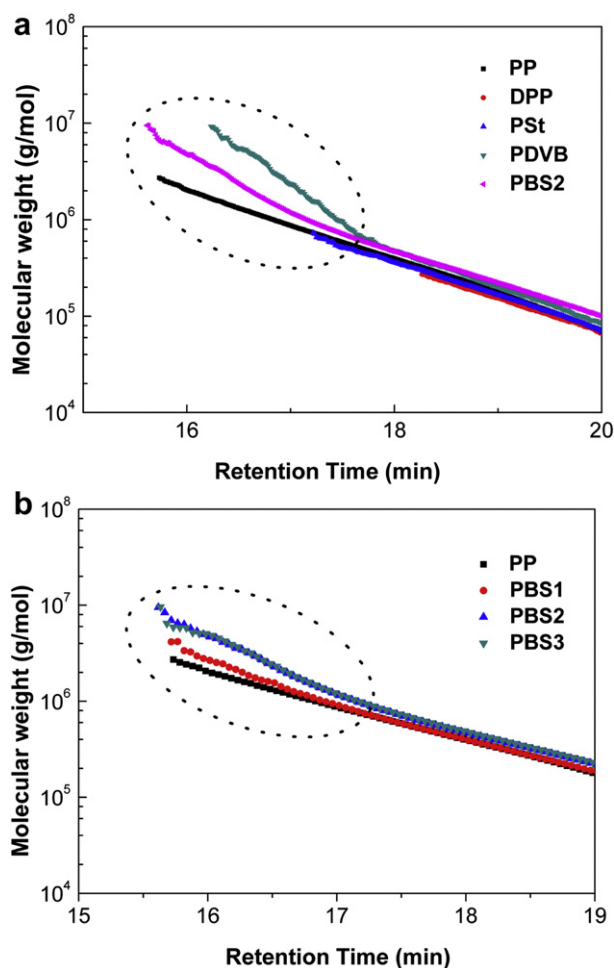


Fig. 3. Molecular weight vs. retention time of PP samples: (a) plain PP, DPP, PSt, PDVB and PBS2; (b) plain PP, PBS1, PBS2 and PBS3.

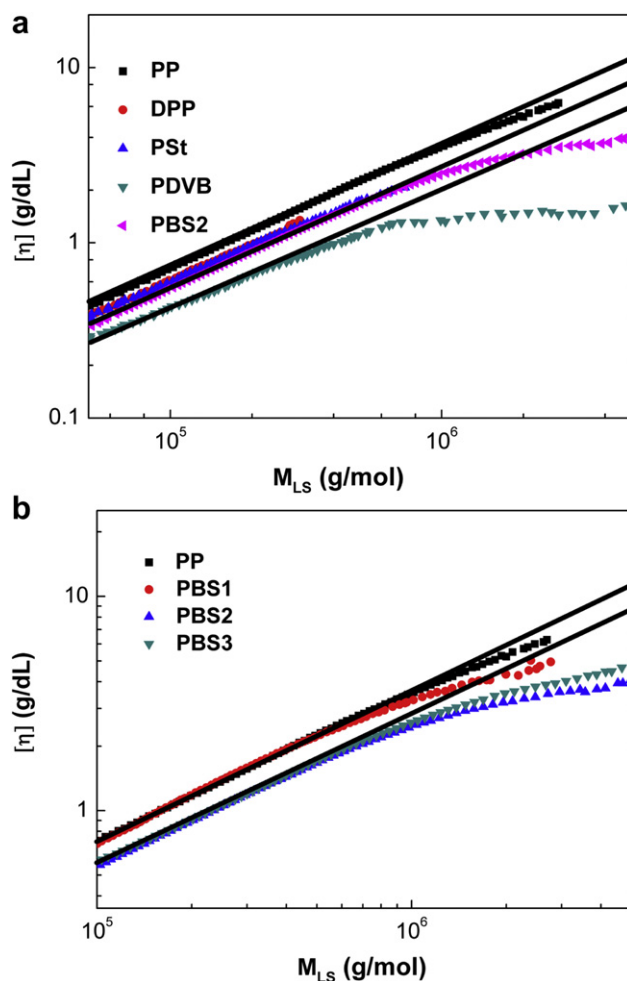


Fig. 4. Mark–Houwink plot of PP samples: (a) plain PP, DPP, PSt, PDVB and PBS2; (b) plain PP, PBS1, PBS2 and PBS3.

Here the suffixes *br* and *lin* represent branched and linear chains. The weight average branching factor (g') can be calculated by Equation (3). It can be seen from Table 1 that the values of g' for PVDB and modified samples with BS are less than one, which confirms the presence of LCB structure on PP backbone [50]. The relationship of g and g' is expressed as:

$$g' = g^\varepsilon \quad (4)$$

Here ε takes values between 0.5 and 1.5 [51]. Based on the reported results, the ε has strong molecular weight dependence [52]. We use $\varepsilon = 0.75$ in this work. For the modified PP, a trifunctional randomly branched architecture can be assumed [53]. Therefore g is expressed in terms of the degree of branching as:

$$g = \left[\left(1 + \frac{m}{7} \right)^{0.5} + \frac{4m}{9\pi} \right]^{-0.5} \quad (5)$$

Here m represents the number of branching points along the molecule. The LCBD is defined as the number of long chain branching points per 10,000 carbons:

$$\text{LCBD} = \frac{m \times 140,000}{M} \quad (6)$$

The values of weight average LCBD for all modified samples are listed in Table 1. Among all the modified samples, PDVB had the highest LCBD. Furthermore, the gel fraction (15 wt%) in PDVB could be regarded as highly branched macromolecule with much higher LCBD. These indicated that the distribution of LCB within PDVB was more non-uniform compared to PBS2 at the same concentration of grafting monomer. Fig. 4b shows the Mark–Houwink plots of PP, PBS1, PBS2 and PBS3. The slope of Mark–Houwink plots for PBS1, PBS2 and PBS3 deviated from linear behavior at high molecular weight. The deviation was more obvious when the concentration of BS was increased. The largest change in slope was observed for PBS2. For PBS1, the lower BS concentration led to the lower LCBD. For PBS3, a part of the grafting monomer was consumed in the gel fraction, so the deviation was not as obvious as PBS2.

Fig. 5a shows molecular weight distribution curves of plain PP, DPP, PSt, PBS2 and PDVB (soluble portion). When only peroxide was loaded, the peak molecular weight shift to low molecular weight. Accordingly, both the peak molecular weight (M_p) and the weight average molecular weight (M_w) of DPP significantly decreased, which resulted from serious chain scission (Table 1). When both peroxide and St were loaded, the shift was smaller. In this case, the chain scission was mild, and the M_p and M_w were 47.4% and 61.2% higher than those of DPP, respectively. These phenomena proved that the degradation of PP was weakened by the addition of St. Compared to plain PP, both PBS2 and PDVB have a tail in the side of high molecular weight due to the presence of LCB. However, the high molecular weight tail of PBS2 was more evident, indicating the more portions of PP macromolecules with LCB. The M_p of PBS2 was lower than that of PP, but the M_w of PBS2 was higher than that of PP. For PBS2, the effect of chain scission on molecular weight was outweighed by the effect of chain branching through BS. For PDVB, while the degradation reduced the molecular weight of the dominant chain population, crosslinking by DVB yielded a minor population of chains with higher LCBD that was less susceptible to molecular weight loss. Thus the M_p and M_w of PDVB are lower than those of plain PP. Considering the formation of gel, PDVB had bimodal molecular weight distribution. Compared to PDVB, PBS2 had more uniform distributions of molecular weight and branching structure. Fig. 5b shows the molecular weight distribution of plain PP and PBS1, PBS2 and PBS3. The peak molecular weight (M_p) and the weight average molecular weight (M_w) were increased with the content of BS, and the tail in higher molecular weight was more

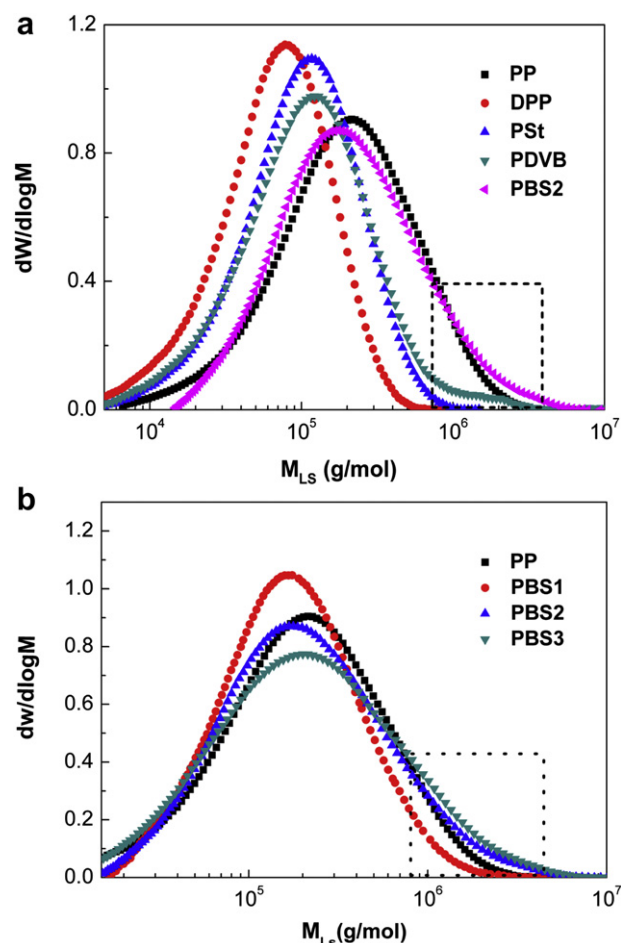


Fig. 5. Molecular weight distribution versus molecular weight of PP samples: (a) plain PP, DPP, PSt, PDVB and PBS2; (b) plain PP, PBS1, PBS2 and PBS3.

evident. The M_p and M_w were listed in Table 1. In the system, the β -scission leads to the decrease of the molecular weight, and the branching reaction results in the increase of the molecular weight. For PBS2 and PBS3, the effect of chain scission on molecular weight was outweighed by the effect of chain branching through BS, which led to the increased of M_w compared to PP. For PBS1, the low LCBD and low LCB fraction resulted in the lower M_w comparing with PP.

Rheological measurements were used to further analyze the microstructure of the resultant samples. Fig. 6 shows the complex viscosity (η^*) versus angular frequency of plain PP and modified PP samples at 180 °C. The rheological behaviors of PP, DPP and PSt are consistent with that of linear polymer. Compared to plain PP, the complex viscosities (η^*) of DPP and PSt decrease, especially in the case of DPP. However, PDVB and the modified samples with BS show strengthened shear thinning at low angular frequency region. Comparatively, PBS2 has a higher η^* than that of PDVB. For the samples modified with various amounts of BS, the η^* was enhanced with increasing BS concentration at low angular frequency region.

Storage modulus (G') at the low frequency, reflecting the long relaxation time, is very sensitive to the topological structure of macromolecules. Fig. 7 shows the storage modulus (G') versus angular frequency for plain PP and modified PP samples at 180 °C. The plain PP, DPP and PSt exhibited the typical terminal behavior, indicating that they are linear polymers. The other samples are deviated from the terminal behavior. G' increased at low angular frequency and the terminal slope of G' decreased. At low angular region, the storage modulus of the sample modified with BS

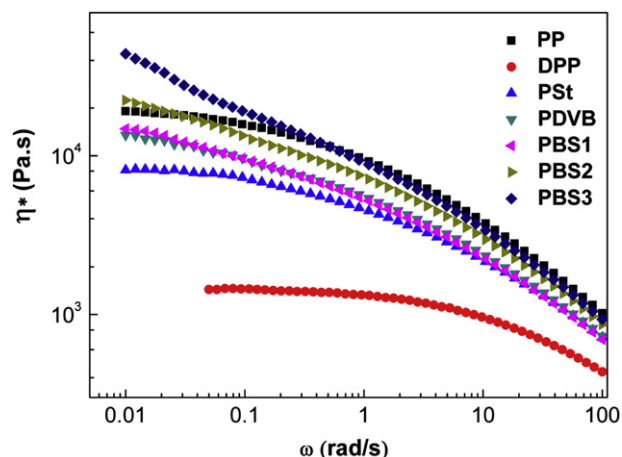


Fig. 6. Complex viscosity (η^*) versus angular frequency of plain PP and modified PP samples at 180 °C.

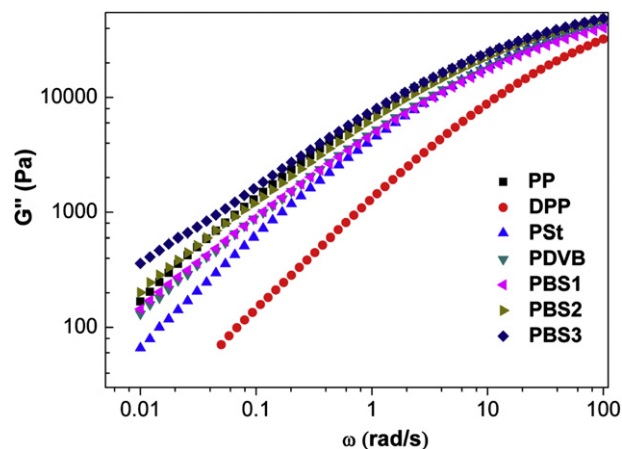


Fig. 8. Loss modulus versus angular frequency of plain PP and modified PP samples at 180 °C.

increased with increasing BS concentration. The non-terminal behavior of the samples modified with BS and PDVB suggest that there is a longer relaxation mechanism, which can be ascribed to the existing of LCB.

Fig. 8 shows the loss modulus (G'') versus angular frequency for plain PP and modified PP samples at 180 °C. The loss modulus (G'') is not sensitive as storage modulus (G') with the LCB structure. Because of the degradation, the G'' of DPP and PSt was lower than that of plain PP in the whole tested angular frequency region. The G'' of PSt was slightly lower than that of plain PP at low frequency region. The G'' increased slightly with increasing BS concentration.

Fig. 9 shows Cole–Cole plot of plain PP and modified PP samples at 180 °C. For plain PP, DPP and PSt, the Cole–Cole plot was close to a semicircle, and the higher the molecular weight was, the bigger the radius was. The samples modified with BS and DVB showed evident upturning at high η' . For the samples modified with BS, the upturning was more evident with the increasing BS concentration.

The phase angle (δ) can be used to evaluate the structures of polymers with LCB by van Gurp–Palmen (vGP) plot [54]. The main idea of the so-called vGP analysis is to plot δ versus the complex modulus (G^*). Fig. 10 shows the vGP plots for all the samples. The data of the plain PP showed the typical shape of a linear polymer. The plot of PSt was almost the same as that of plain PP. The molecular weight of DPP was much lower than that of plain PP,

therefore the entanglement point of DPP was much lower than that of plain PP, indicating that DPP was less elastic (more viscous). Thus, the δ of DPP was higher at the same values of G^* compared to plain PP. The values of δ for PDVB and PBS2 samples were lower than that of the plain PP at lower values of G^* . The values of δ decreased with increasing BS concentration at lower G^* , which indicated the presence of more elastic portion in the melt. The deviations were related to the existence of LCB in these samples [55].

Elongational flow experiment is a sensitive tool for the investigation of LCB. Fig. 11 shows extensional viscosity of plain PP, PDVB, PBS1, PBS2 and PBS3 measured at 180 °C. For plain PP, no strain hardening was observed. The samples modified with PDVB and BS showed a strain hardening. However, the sample modified with BS (PBS2) had higher melt strength (MS) than that of PDVB at the same concentration of grafting monomer (Table 1). For the samples modified with BS, the strain hardening became more pronounced and the MS became higher with increasing BS concentration (Table 1). These improved rheological properties result from more uniform distribution of LCB in PP backbone and the mild degradation of PP chains for the samples modified with BS compared to PDVB.

The mixing torque curves of PP samples are shown in Fig. 12. For plain PP, the torque curve kept nearly horizontal after complete melting with the end-torque value of 8.8 N m. When only peroxide was added (DPP), the mixing torque value decreased gradually until

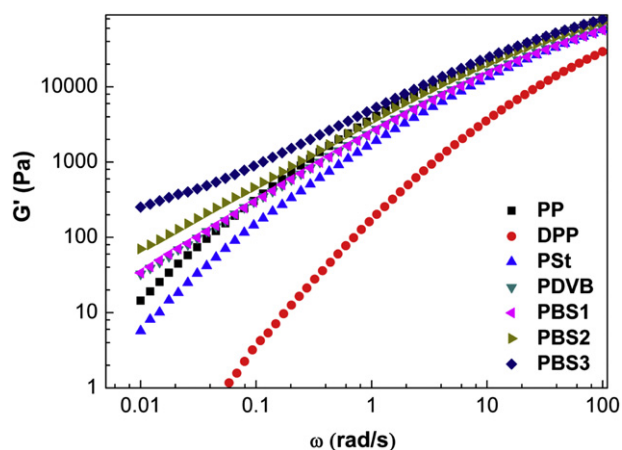


Fig. 7. Storage modulus versus angular frequency of plain PP and modified PP samples at 180 °C.

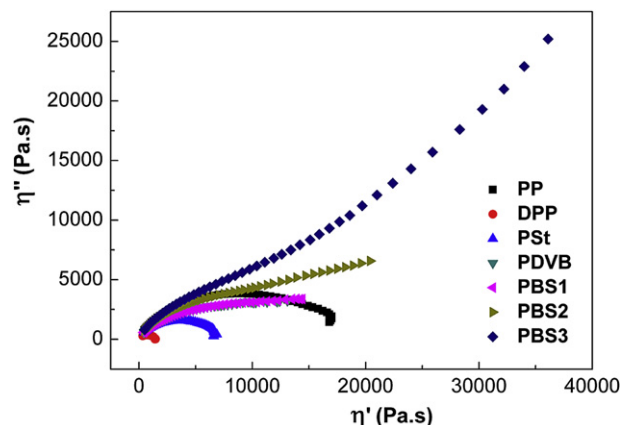


Fig. 9. Cole–Cole plots for plain PP and modified PP samples at 180 °C.

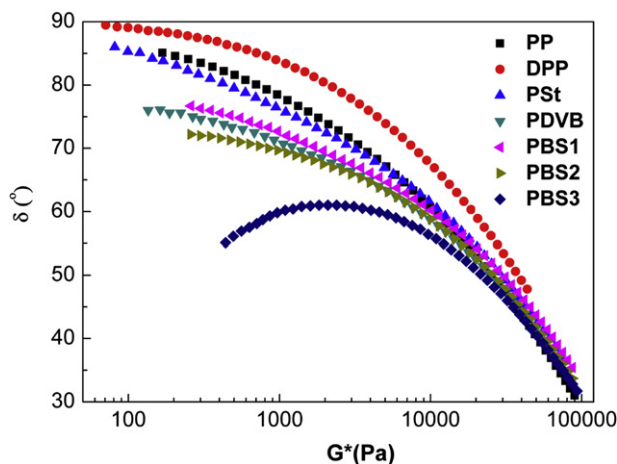


Fig. 10. Van Gurp–Palmen plots for plain PP and modified PP samples.

the end of reaction with the end-torque value of 3.8 N m. However, when both peroxide and styrene (St) were added (PSt), the end-torque value was higher than that of DPP. The reason is that the PP macroradicals were changed to styryl macroradicals by reacting with St, thus the β -scission of PP macroradicals could be restrained. When divinylbenzene (DVB) was used as a grafting monomer, a notable second torque peak appeared, and then the value of the torque decreased gradually with the end-torque value of 7.5 N m that was lower than that of plain PP. These phenomena were also observed by other researchers [56]. The high reactivity double-bond in DVB resulted in uncontrolled branching/crosslinking corresponding to the second torque peak. However, after the DVB consumed quickly, the residual peroxide will lead to the chain scission corresponding to the slow decrease of the torque at late reactive stage. When BS was used as a grafting monomer, there was no second torque peak and the end-torque value was higher than that of plain PP except for PBS1. This is because the high reactive double-bond in BS firstly reacted with the PP macroradical to form stable radical to restrain the degradation of PP just like St, and then the lower reactive double-bond in BS gradually reacted with PP macroradical to form branched or light crosslinked structure to increase the molecular weight. The low reaction rate can also be confirmed from ^1H NMR spectra of purified PBS2. Fig. 13 shows that there are still some lower reactivity double-bonds (δ 5.1 ppm

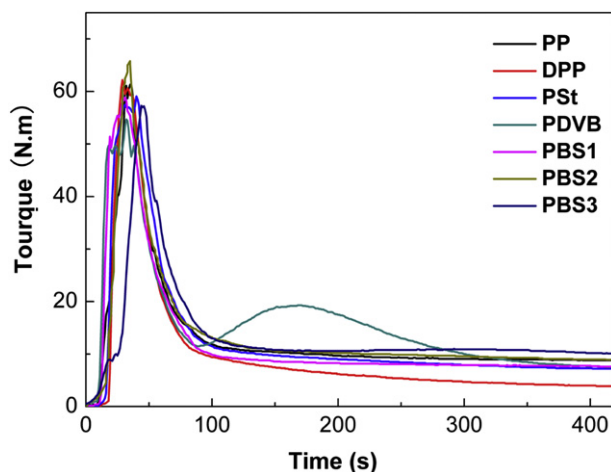


Fig. 12. Torque behavior of plain PP and modified PP samples.

($\text{C}=\text{CH}_2$)) while there is no higher reactivity double-bond (δ 5.3 ppm (aromatic- $\text{C}=\text{CH}$)) in the sample of PBS2.

^1H NMR analysis of unpurified modified PP samples also provided the information about the conversion of grafting monomer. Fig. 14 shows the ^1H NMR spectra of unpurified PBS2 (a) and PBS3 (b) in tertachloroethane- d_2 at 110 °C. The chemical shifts at 7.1 ppm and 7.4 ppm correspond to the hydrogen resonance in benzene. The chemical shift at 6.71 ppm in Fig. 14b corresponds to the hydrogen resonance in the aromatic $-\text{CH}=\text{C}$ which originates from the unreacted BS monomer. The chemical shift at 6.71 ppm was not detected in the spectra of PBS2 (Fig. 14a), PBS1 (not shown), PSt (not shown) and PDVB (not shown), indicating that the conversion of the grafting monomers in these samples were nearly 100%. The conversion of BS in PBS3 sample was about 70%.

Fig. 15 shows schematic drawing of chemical reaction between PP macroradical and various grafting monomers. The chemical structures of grafting monomers play a key role in mediating the melt reaction of PP. Although three kinds of grafting monomer can stabilize PP macroradicals through the formation of styryl macroradicals, their influences on the subsequent reaction are different. It is difficult to form LCB structure through the reaction of the intermediate radicals with PP macroradicals due to steric hindrance. When the grafting monomer is styrene, no LCB structure is formed on the PP backbone. When the grafting monomer is DVB, the other double-bond on the grafted DVB is still high reactive to radical species. In this case, the crosslinking reaction will take place

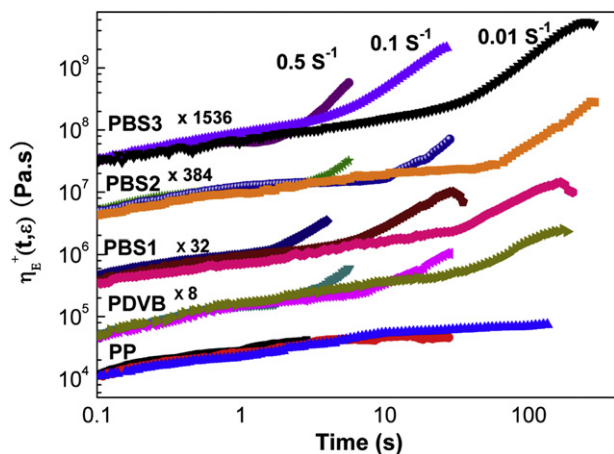


Fig. 11. Extensional viscosities for plain PP, PDVB, PBS1, PBS2 and PBS3 measured at 180 °C.

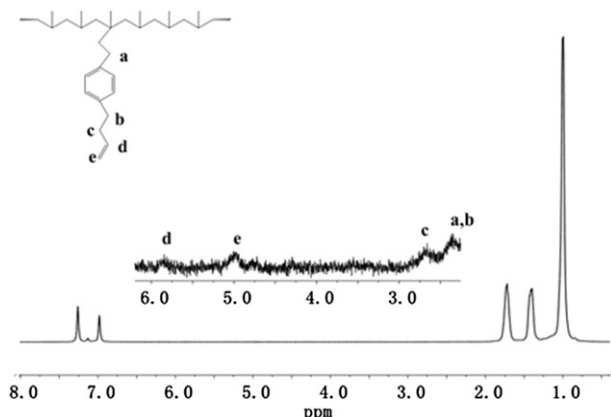


Fig. 13. ^1H NMR spectra of purified PBS in tertachloroethane- d_2 at 110 °C.

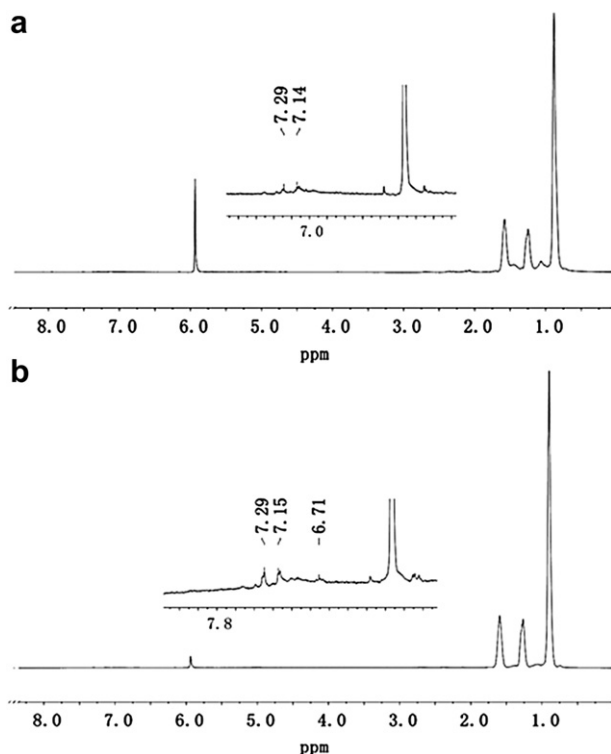


Fig. 14. ^1H NMR spectra of unpurified PBS2 (a) and PBS3 (b) in tertachloroethane- d_2 at 110 °C.

besides the branching reaction. According to Parent's results [34], grafting monomer is preferentially grafted to high molecular weight chains, which in turn will have a heightened capacity for crosslinking through further additions to bound C=C functionality. Therefore, the monomer of DVB was prone to resulting in uncontrolled branching/crosslinking reaction, which was confirmed by the formation of an amount of gel and the second mixing torque peak. When the grafting monomer is BS, the reactivity of two double-bonds is totally different. The high reactive double-bond in BS firstly reacts with the PP macroradical to form stable radical, and then the lower reactive double-bond reacts with PP macroradical to form branched structure. Clearly the BS monomer shows double function in the system, one of them stabilizing PP macroradicals and the other promoting branching reaction. Compared to PDVB,

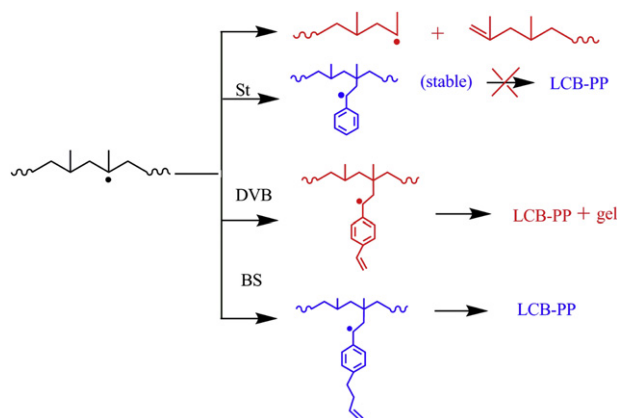


Fig. 15. Schematic drawing of chemical reaction between PP macroradical and grafting monomers.

Table 2

Mechanical and crystallization properties of plain PP and modified PP samples.

Sample	σ_m (MPa)	E_t (MPa)	T_m (°C)	T_c (°C)	ΔH_f (J/g)	X_c (%)
PP	36.0	820	163.8	115.2	100.7	48.2
DPP	35.1	950	163.2	117.6	103.6	49.6
PSt	35.8	1030	163.9	117.0	101.6	48.6
PDVB	38.7	930	165.1	129.6	107.5	51.4
PBS1	36.5	900	164.7	124.3	104.7	50.1
PBS2	36.6	980	165.0	126.1	106.8	51.1
PBS3	36.9	1050	165.4	129.5	102.7	49.1

PBS shows more uniform distributions of molecular weight and branching structures.

The mechanical properties and crystallization properties of the modified samples were also studied in this work. The mechanical properties of plain PP and modified samples were listed in Table 2. DPP and PSt showed a little higher Young modulus (E_t) but a little lower tensile strength (σ_m) compared to plain PP. However, the LCB-PP samples (PDVB, PBS1, PBS2 and PBS3) had both higher E_t and σ_m , which is related to the self nucleation by LCB or the fraction having an extremely high molecular weight in branched part [17]. The cooling and heating thermograms of plain PP and modified PP samples are shown in Fig. 16, and the corresponding crystallization and melting parameters are listed in Table 2. It can be seen from

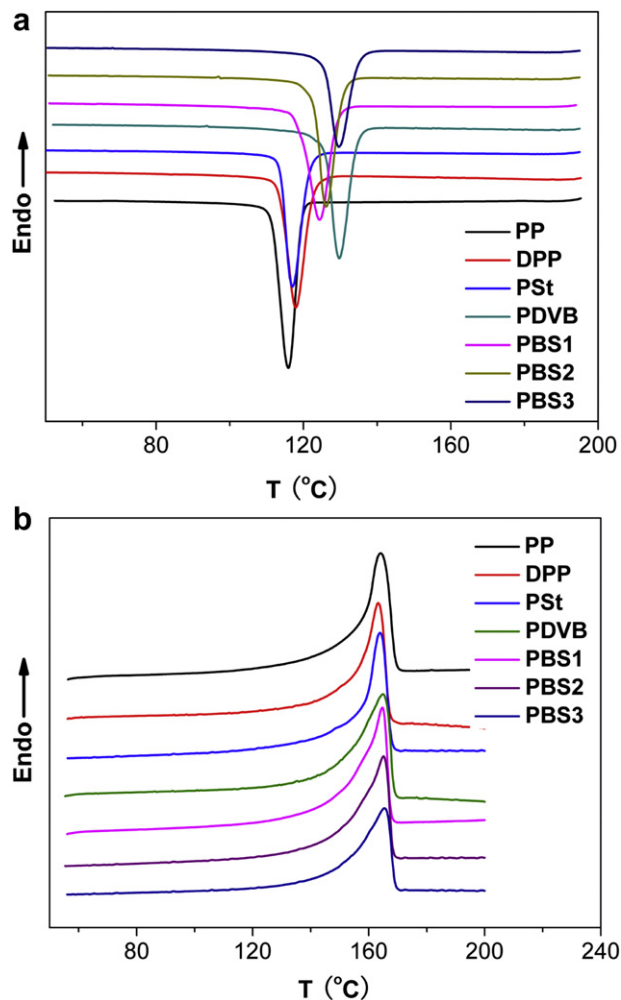


Fig. 16. DSC cooling thermograms (a) and melting thermograms (b) for plain PP and modified samples at 10 °C/min.

cooling curves in Fig. 16a that crystallization temperature (T_c) of LCB samples is much higher than that of PP. As shown in Table 2, T_c of PP is 115.2 °C and T_c of PDVB, PBS1, PBS2 and PBS3 is 129.6, 124.3, 126.1 and 129.5 °C, respectively, while the T_c of DPP and PSt is little higher than that of PP. It is clear that the presence of LCB and gel increases the T_c . The reason is that the presence of LCB and the gel fraction promoted heterogeneous nucleation and accelerated the crystallization process of PP chains [57,58]. Fig. 16b shows the second heating thermograms of PP and modified samples. The melt temperature (T_m) of the samples modified with DVB and BS shifted to a little higher temperature comparing with linear PP. The crystallinity of PP samples is listed in Table 2. It can be seen that the crystallinity of the samples modified with DVB and BS is a little higher than that of linear PP, indicating that LCB helps to increase the crystallization ability of PP.

4. Conclusion

Comparing DVB with BS, the reactivity ratio of two double-bonds in a grafting monomer to macroradicals is a key factor to control melt reaction. BS containing two double-bonds with different reactivity has the ability to mediate melt radical reaction. The high reactive double-bond in BS preferentially react with PP macroradicals to form stable radical and restrain the degradation of PP; the low reactive double-bond in the grafted BS could further react with PP macroradical to form branched structure due to the low reaction rate. Compared to DVB, the addition of BS depressed the crosslinking reaction. Thus using BS as a grafting monomer results in the formation of more uniform LCB structure on the PP backbone. The introduction of LCB to PP backbone improved slightly mechanical properties and strongly change the crystallization behaviors. It is believed that further optimizing chemical structure of grafting monomer, especially the reactivity ratio of two double-bonds in grafting monomer, will be beneficial to mediating melt reaction.

Acknowledgments

We thank the financial supports from the National Natural Science Foundation of China for the projects (20734006, 20804045, 50873099 and 50921062) and the Ministry of Science and Technology of China for Project No. 2008AA03Z508.

References

- [1] Kraus G, Gruver J. *J Polym Sci Part A Polym Chem* 1965;3(1):105–22.
- [2] Raju V, Smith G, Marin G, Knox J, Graessley W. *J Polym Sci Part B Polym Phys* 1979;17(7):1183–95.
- [3] Bersted B, Slee J, Richter C. *J Appl Polym Sci* 1981;26(3):1001–14.
- [4] Graessley W. *Macromolecules* 1982;15(4):1164–7.
- [5] Shroff R, Prasad A, Lee C. *J Polym Sci Part B Polym Phys* 1996;34(14):2317–33.
- [6] Gahleitner M. *Prog Polym Sci* 2001;26(6):895–944.
- [7] Roovers J. *Macromolecules* 1984;17(6):1196–200.
- [8] He C, Costeux S, Wood-Adams P, Dealy JM. *Polymer* 2003;44(23):7181–8.
- [9] Wood-Adams P, Dealy J, Willem deGroot A, Redwine O. *Macromolecules* 2000;33(20):7489–99.
- [10] Soares JBP. *Macromol Mater Eng* 2004;289(1):70–87.
- [11] Malmberg A, Gabriel C, Steffl T, Münstedt H, Lofgren B. *Macromolecules* 2002;35(3):1038–48.
- [12] Stadler F, Kaschta J, Münstedt H, Becker F, Buback M. *Rheol Acta* 2009;48(5):479–90.
- [13] Kessner U, Kaschta J, Stadler FJ, Le Duff CS, Drooghaag X, Münstedt H. *Macromolecules* 2010;43(17):7341–50.
- [14] Kempf M, Barroso VC, Wilhelm M. *Macromol Rapid Commun* 2010;31(24):2140–5.
- [15] Cheng S, Phillips E, Parks L. *Radiat Phys Chem* 2010;79(3):329–34.
- [16] McCallum T, Kontopoulou M, Park C, Muliawan E, Hatzikiriakos S. *Polym Eng Sci* 2007;47(7):1133–40.
- [17] Rätzsch M, Arnold M, Borsig E, Bucka H, Reichelt N. *Prog Polym Sci* 2002;27(7):1195–282.
- [18] Auhl D, Stange J, Münstedt H, Krause B, Voigt D, Lederer A, et al. *Macromolecules* 2004;37(25):9465–72.
- [19] Gotsis A, Zeevenhoven B, Hogt A. *Polym Eng Sci* 2004;44(5):973–82.
- [20] Sugimoto M, Suzuki Y, Hyun K, Hyun Ahn K, Ushioda T, Nishioka A, et al. *Rheol Acta* 2006;46(1):33–44.
- [21] Spital P, Macosko CW. *Polym Eng Sci* 2004;44(11):2090–100.
- [22] Lagendijk RP, Hogt AH, Buijtenhuijs A, Gotsis AD. *Polymer* 2001;42(25):10035–43.
- [23] Ye Z, Alobaidi F, Zhu S. *Ind Eng Chem Res* 2004;43(11):2860–70.
- [24] Wang WJ, Ye ZB, Fan H, Li BG, Zhu SP. *Polymer* 2004;45(16):5497–504.
- [25] Weng W, Hu W, Dekmezian AH, Ruff CJ. *Macromolecules* 2002;35(10):3838–43.
- [26] Langston JA, Colby RH, Chung TCM, Shimizu F, Suzuki T, Aoki M. *Macromolecules* 2007;40(8):2712–20.
- [27] Wang L, Wan D, Zhang Z, Liu F, Xing H, Wang Y, et al. *Macromolecules* 2011;44(11):4167–79.
- [28] Krause B, Voigt D, Haussler L, Auhl D, Münstedt H. *J Appl Polym Sci* 2006;100(4):2770–80.
- [29] Schulze D, Trinkle S, Mülhaupt R, Friedrich C. *Rheol Acta* 2003;42(3):251–8.
- [30] Lugaõ AB, Otaguro H, Parra DF, Yoshiga A, Lima L, Artel BWH, et al. *Radiat Phys Chem* 2007;76(11–12):1688–90.
- [31] Tian J, Yu W, Zhou C. *Polymer* 2006;47(23):7962–9.
- [32] Li S, Xiao M, Wei D, Xiao H, Hu F, Zheng A. *Polymer* 2009;50(25):6121–8.
- [33] Mabrouk EK, Parent JS, Chaudhary BI, Cong RJ. *Polymer* 2009;50(23):5390–7.
- [34] Parent JS, Sengupta SS, Kaufman M, Chaudhary BI. *Polymer* 2008;49(18):3884–91.
- [35] Graebing D. *Macromolecules* 2002;35(12):4602–10.
- [36] Zhang Z, Xing H, Qiu J, Jiang Z, Yu H, Du X, et al. *Polymer* 2010;51(7):1593–8.
- [37] Zhang Z, Wan D, An Y, Liu F, Xing H, Wang L, et al. *Polym Degrad Stabil* 2011;96(4):653–9.
- [38] Cartier H, Hu GH. *J Polym Sci Part A Polym Chem* 1998;36:1053–63.
- [39] Li Y, Xie XM, Guo BH. *Polymer* 2001;42(8):3419–25.
- [40] Coiai S, Passaglia E, Aglietto M, Ciardelli F. *Macromolecules* 2004;37(22):8414–23.
- [41] Coiai S, Augier S, Pinzino C, Passaglia E. *Polym Degrad Stabil* 2010;95(3):298–305.
- [42] Ma J, Cheng C, Sun G, Wooley K. *Macromolecules* 2008;41(23):9080–9.
- [43] Dong Z, Liu X, Liu H, Li Y. *Macromolecules* 2010;43(19):7985–92.
- [44] Langston JA, Colby RH, Shimizu F, Suzuki T, Aoki M, Chung TCM. *Macromol Symp* 2007;260(1):34–41.
- [45] Dong JY, Manias E, Chung TC. *Macromolecules* 2002;35(9):3439–47.
- [46] Arroyo M, Lopez-Manchado MA, Avalos F. *Polymer* 1997;38(22):5587–93.
- [47] Beer F, Capaccio G, Rose L. *J Appl Polym Sci* 1999;73(14):2807–12.
- [48] Yau W, Gillespie D. *Polymer* 2001;42(21):8947–58.
- [49] Sun T, Brant P, Chance RR, Graessley WW. *Macromolecules* 2001;34(19):6812–20.
- [50] Zimm BH, Stockmayer WH. *J Chem Phys* 1949;17(12):1301–14.
- [51] Jørgensen JK, Redford K, Ommundsen E. *J Appl Polym Sci* 2007;106(2):950–60.
- [52] Wang WJ, Kharchenko S, Migler K, Zhu SP. *Polymer* 2004;45(19):6495–505.
- [53] Krause B, Voigt D, Lederer A, Auhl D, Münstedt H. *J Chromatogr A* 2004;1056(1–2):217–22.
- [54] Trinkle S, Walter P, Friedrich C. *Rheol Acta* 2002;41(1):103–13.
- [55] Lohse DJ, Milner ST, Fetters LJ, Xenidou M, Hadjichristidis N, Mendelson RA, et al. *Macromolecules* 2002;35(8):3066–75.
- [56] Romani F, Corrieri R, Braga V, Ciardelli F. *Polymer* 2002;43(4):1115–31.
- [57] Zeng W, Liu J, Zhou J, Dong JL, Yan SK. *Chin Sci Bull* 2007;52(19):2246–54.
- [58] Tian J, Yu W, Zhou C. *J Appl Polym Sci* 2007;104(6):3592–600.